

$$\begin{aligned}
||G_{\tilde{z}e}||_{\infty} &= \sup_{\omega} \frac{||\tilde{z}||_2}{||e||_2} \\
&= \sup_{\omega} \sigma_1[G_{\tilde{z}e}(j\omega)]
\end{aligned} \tag{11.126}$$

where $G_{\tilde{z}e}(s)$ is the transfer function from e to $\tilde{z}$.

11.5 TRANSFER FUNCTION APPROACHES

It should be emphasized that other formulations to H_{∞} filtering have been proposed. For instance, Isaac Yaesh and Uri Shaked [Yae91] consider the following time-invariant system:

$$\begin{aligned}
x_{k+1} &= Fx_k + w_k \\
x_0 &= 0 \\
y_k &= Hx_k + v_k \\
z_k &= Lx_k
\end{aligned} \tag{11.127}$$

where w_k and v_k are uncorrelated process and measurement noise, y_k is the measurement, and z_k is the vector to be estimated. Define the estimation error as

$$\tilde{z}_k = z_k - \hat{z}_k \tag{11.128}$$

Define an augmented disturbance vector as

$$e_k = \begin{bmatrix} w_k \\ v_k \end{bmatrix} \tag{11.129}$$

The goal is to find a steady-state estimator such that the infinity-norm of the transfer function from the augmented disturbance vector e to the estimation error $\tilde{z}$ is less than some user specified bound:

$$||G_{\tilde{z}e}||_{\infty}^2 < \frac{1}{\theta} \tag{11.130}$$

The steady-state *a priori* filter that solves this problem is given as

$$\begin{aligned}
P &= I + FPF^T - FPH^T(I + HPH^T)^{-1}HPF^T + \\
&\quad PL(I/\theta + LPL^T)^{-1}LP \\
K &= FPH^T(I + HPH^T)^{-1} \\
\hat{x}_{k+1} &= F\hat{x}_k + K(y_k - H\hat{x}_k)
\end{aligned} \tag{11.131}$$

These equations solve the H_{∞} estimation problem if and only if P is positive definite.

The steady-state *a posteriori* filter that solves this problem is given as

$$\begin{aligned}
\Sigma^{-1} &= \tilde{P}^{-1} - \theta L^T L + H^T H \\
\tilde{P} &= F\tilde{P}(H^T H \tilde{P} - \theta L^T L \tilde{P} + I)^{-1}F^T + I \\
\tilde{K} &= (I + \theta L^T L)^{-1}\Sigma H^T \\
&= \tilde{P}(I + H^T H \tilde{P})^{-1}H^T \\
\hat{x}_{k+1} &= F\hat{x}_k + \tilde{K}(y_{k+1} - HF\hat{x}_k)
\end{aligned} \tag{11.132}$$

Again, these equations solve the H_∞ estimation problem if and only if $\tilde{P}$ is positive definite.

Interestingly, the P matrix in the *a priori* filter of Equation (11.131) is related to the $\tilde{P}$ matrix in the *a posteriori* filter of Equation (11.132) by the following equation:

$$P^{-1} = \tilde{P}^{-1} - \theta L^T L \quad (11.133)$$

In general, the Riccati equations in these filters can be difficult to solve. However, the solution can be obtained by the eigenvector method shown in [Yae91]. (This is similar to the Hamiltonian approach to steady-state Kalman filtering described in Section 7.3.3.) Define the $2n \times 2n$ matrix

$$\mathcal{H} = \begin{bmatrix} F^T + H^T H F^{-1} & \theta F^T L^T L - H^T H F^{-1}(I - \theta L^T L) \\ -F^{-1} & F^{-1}(I - \theta L^T L) \end{bmatrix} \quad (11.134)$$

Note that F^{-1} should always exist if it comes from a real system, because F comes from a matrix exponential that is always invertible (see Sections 1.2 and 1.4). Compute the n eigenvectors of $\mathcal{H}$ that correspond to the eigenvalues outside the unit circle. Denote those eigenvectors as ξ_i ($i = 1, \dots, n$). Form the $2n \times n$ matrix

$$\begin{bmatrix} \xi_1 & \cdots & \xi_n \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \quad (11.135)$$

where X_1 and X_2 are $n \times n$ matrices. The P matrix used in the *a priori* H_∞ filter can be computed as

$$P = X_2 X_1^{-1} \quad (11.136)$$

For the *a posteriori* filter, define the $2n \times 2n$ matrix

$$\tilde{\mathcal{H}} = \begin{bmatrix} F^{-T} & F^{-T}(H^T H - \theta L^T L) \\ F^{-T} & F + F^{-T}(H^T H - \theta L^T L) \end{bmatrix} \quad (11.137)$$

Compute the n eigenvectors of $\tilde{\mathcal{H}}$ that correspond to the eigenvalues outside the unit circle. Denote those eigenvectors as $\tilde{\xi}_i$ ($i = 1, \dots, n$). Form the $2n \times n$ matrix

$$\begin{bmatrix} \tilde{\xi}_1 & \cdots & \tilde{\xi}_n \end{bmatrix} = \begin{bmatrix} \tilde{X}_1 \\ \tilde{X}_2 \end{bmatrix} \quad (11.138)$$

where $\tilde{X}_1$ and $\tilde{X}_2$ are $n \times n$ matrices. The $\tilde{P}$ matrix used in the *a posteriori* H_∞ filter can be computed as

$$\tilde{P} = \tilde{X}_2 \tilde{X}_1^{-1} \quad (11.139)$$

The eigenvector method for the Riccati equation solutions works because $\mathcal{H}$ and $\tilde{\mathcal{H}}$ are symplectic matrices (see Section 7.3.3 and Problem 11.9). This assumes that F is nonsingular and that $\mathcal{H}$ and $\tilde{\mathcal{H}}$ do not have any eigenvalues on the unit circle. If these assumptions are violated, then the problem becomes more complicated [Yae91]. A method similar to this for continuous-time systems is developed in [Nag91].

It is important to be aware that the P and $\tilde{P}$ solutions given by Equations (11.136) and (11.139) only give one solution each to Equations (11.131) and (11.132). Equations (11.136) and (11.139) may give solutions to Equations (11.131) and (11.132) that are not positive definite and therefore do not satisfy the H_∞ filtering problem. However, that does not prove that the H_∞ filtering solution does not exist (see Problem 11.13).